SANTHOSH P

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EDUCATION

Karpagam Institute of Technology **2024 – Present**
B.E. Computer Science Engineering - ***CGPA - 8.72*** *Coimbatore, India*

Jayanthi Matriculation Higher Secondary School **2022 – 2024**
HSC (12th Grade) - ***Percentage - 91%*** *Coimbatore, India*

COURSEWORK / SKILLS

- Data Structures & Algorithms
- Artificial Intelligence
- Operating Systems
- Computer Networks
- System Design
- Database Management Systems
- Object-Oriented Programming
- Web Development

PROJECTS

Deepfake-Agentic-AI | *Python, FastAPI, Docker, LangChain, LangGraph, PostgreSQL* **Nov 2025 – Jun 2026**
- Built an explainable agentic deepfake detection pipeline (OpenCV, custom CNNs, Vision Transformers) achieving **94% accuracy** on a self-labeled public image test set
- Integrated an LLM reasoning layer (Gemma for log parsing, LLaMA for orchestration) to generate human-readable explanations for every detection, closing the black-box interpretability gap common in deepfake detection systems
- Engineered a containerized microservice architecture (Docker networks, MinIO, PostgreSQL) with safety guardrails and real-time observability (Dozzle, Beszel)
- Validated the pipeline with **40+ automated test suites** and versioned Docker artifacts across the CI/CD pipeline

Multimodal RAG Assistant | *Python, FastAPI, LangGraph, Next.js, MCP* **Apr 2026 – Sept 2026**
- Architected a multimodal RAG system (SigLIP + all-MiniLM-L6-v2 embeddings, adaptive chunking, hybrid search) to automate Tier-1 support document retrieval, achieving **98% retrieval accuracy**, **100% hit rate**, and **84% P95** across recall@3/recall@5 benchmarks
- Optimized vector similarity search using dot-product on L2-normalized embeddings in place of cosine similarity, eliminating redundant division/sqrt operations across millions of comparisons at scale (500+ chunks/document)
- Built a 5-layer caching system with perceptual hashing (pHash/mHash) for deduplication and async parallel query pipelines; deployed as a local MCP server integrated with Claude Desktop, validated across **42 test suites**

doc2site | *JavaScript, GitHub Actions, Static Site Generation* **Dec 2025 – Jan 2026**
- Built an automated static-site deployment engine converting Markdown docs into live, custom-styled sites via a single repo push, no manual deployment or hosting setup required
- Powers **15+ live documentation sites** to date, fully driven by GitHub Actions with support for custom CSS theming

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, Rust, R
AI/ML: Hugging Face Transformers, Ollama, vLLM, MLflow, LangChain, LangGraph
Databases: PostgreSQL, Qdrant, pgvector, MongoDB
Frameworks: FastAPI, React, Next.js, Node.js
DevOps & Tools: Docker, Podman, GitHub Actions, Git, Linux, Dozzle, Beszel
Cloud & Data: AWS, Excel (large-scale data analysis), Matplotlib, D2, Mermaid

EXTRACURRICULAR

Leadership & Community Involvement **2024 – 2026**
CloudWeld, UG Madurai, Techtronautz *Karpagam Institute of Technology*
- **CloudWeld (Cloud Club):** progressed from Member (2nd year) to **Technical Lead** (3rd year, current), driving cloud-focused technical sessions
- **UG Madurai Community:** ranked **#1** on the college-club leaderboard for **2+ months**; organized AWSBlocks (AWS event, **72 participants**) under UG Sprint; regular at build sessions and **20+ workshops** on frontier tech
- **Techtronautz (Dept. Club):** co-organized **7–8 events** with seniors and club leaders (2nd year), building leadership and public-speaking skills

Competitive Programming & Hackathons **2025 – 2026**
Team Lead *LeetCode, CodeChef, ICPC*
- Solved **500+ LeetCode** and **200+ CodeChef** problems; competed in **25+ contests**; registered for **ICPC 2026–27**
- Led diverse teams across **50+ hackathons**; ranked **Top 10** on the UG Builder Circle leaderboard